

The empirical recurrence for OEIS A283947: recovering a conjecture that is not written down

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Abstract

OEIS A283947 records “Empirical recurrence of order 90”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 150$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 90, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 90 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283947 is “Number of nX5 0..1 arrays with no 1 equal to more than three of its horizontal, diagonal and antidiagonal neighbors, with the exception of exactly one element.”. It has offset 1 and begins

0, 84, 7988, 264482, 8244557, 242330064, 6816984648, 186634970946, ...

The entry states:

Empirical recurrence of order 90 (see link above)

As of the “Last modified” line on the live entry (Mar 18 2017 12:22:45, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1908$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 150$ of the 1908, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 150$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 300$ exact terms of the model returns order 90 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{90} c_i a(n-i),$$

c_1	46	c_{24}	31603914040414416	c_{47}	-214314691307017121064	c_{70}	-714826733710
c_2	-569	c_{25}	4934423672303446	c_{48}	113785724503214364541	c_{71}	1013874944503
c_3	2662	c_{26}	22349308235566685	c_{49}	405823620271365564944	c_{72}	221757609364
c_4	-46898	c_{27}	-158365845217534190	c_{50}	-103429960757543828446	c_{73}	-30395295933
c_5	207596	c_{28}	-331536106407224828	c_{51}	-552393458495089497688	c_{74}	-48854029066
c_6	-1850643	c_{29}	2393816398366247904	c_{52}	146008570445819956740	c_{75}	141572581118
c_7	15162664	c_{30}	-5085338820689039928	c_{53}	372093684795346521926	c_{76}	17158090585
c_8	-40873076	c_{31}	6163930639316863100	c_{54}	10214424919843786295	c_{77}	37384474282
c_9	448701066	c_{32}	-8292486861304857579	c_{55}	-361490265195131317022	c_{78}	-8002624664
c_{10}	-1939313428	c_{33}	14388392024523052542	c_{56}	-15320242559971644382	c_{79}	51478956630
c_{11}	5564651232	c_{34}	-13641753380008633455	c_{57}	271895029260845258216	c_{80}	71166544120
c_{12}	-56897104781	c_{35}	-7873423664259810252	c_{58}	-63467813803533178829	c_{81}	-1484450932
c_{13}	80522627016	c_{36}	26504026569818622428	c_{59}	-129163389757457704792	c_{82}	3838054896
c_{14}	-258640197691	c_{37}	-19225626794808895048	c_{60}	41124823386043750874	c_{83}	-171229044
c_{15}	4055901402292	c_{38}	2585224303934906650	c_{61}	86801986923752325662	c_{84}	103848724
c_{16}	-130819500543	c_{39}	-89797619562483771154	c_{62}	-25939005481662635665	c_{85}	939111178
c_{17}	-8864627955716	c_{40}	101452731708214281594	c_{63}	-54727022887849524370	c_{86}	-13882757
c_{18}	-129806186096149	c_{41}	14362988252926100304	c_{64}	32275989698608532364	c_{87}	43757076
c_{19}	-32144040264614	c_{42}	-76976086021359189106	c_{65}	13868892281796377618	c_{88}	-9994240
c_{20}	933656281191558	c_{43}	-71284816835226161784	c_{66}	-18911702918130367212	c_{89}	31195130
c_{21}	697707524851630	c_{44}	42331898921335473360	c_{67}	1735494752594570432	c_{90}	-262144
c_{22}	216390947267763	c_{45}	290183555746370256956	c_{68}	4697344967289295511		
c_{23}	-19819853134413808	c_{46}	-282633671838303147394	c_{69}	-1210722454866663112		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 90, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $90 + 4$ terms removed returns order 90 as well, so $\deg q_{\min} = 90 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $90 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 90 that this sequence satisfies — and that family turns out to have exactly one member.

References

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